

BTFR Zero Point from Cosmological Horizon Memory

Hysterical Response derivation of $a_0 = H_0 c / 2\pi$

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Computational note. Proofs, exploratory experiments, and paper writing were assisted by generative AI. Mathematical theorems stated in this paper are formally verified in Lean 4 in the companion specifications, as faithful ports of the TeX arguments at the abstractions recorded there, except results explicitly tagged as *literature hypotheses* (standard theorems cited from the literature and not re-proved here) or *physical inputs* (modeling assumptions outside mathematics).

Abstract

The galactic-modelling companion derives the baryonic Tully–Fisher (BTFR) exponent $v_f^4 \propto M_b$ from a capacity attractor, but leaves the absolute acceleration scale a_0 encoded in a combination of response couplings. This paper closes that zero point. Hysterical Response is causal memory of an already-existing interaction; the softest gravitational residual in an expanding universe is IR-pinned by cosmological-horizon KMS memory with Euclidean period $\beta = 2\pi/H_0$. The associated memory acceleration is $a_{\text{IR}} = c/\beta = H_0 c / 2\pi$. Matching the mature BTFR writing $v_f^4 = G a_0 M_b$ to that scale yields

$$a_0 = \frac{H_0 c}{2\pi}.$$

The factor 2π is horizon regularity, not a MOND fitting constant. The scale is a property of the residual gravitational channel and the FLRW congruence, not of any particular probe’s age. No new force carrier is introduced: the result normalizes ordinary gravity’s Hysterical leftover, organized as the galactic halo of the companion paper.

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1 Scope and status

Claim. On the mature BTFR attractor of the galactic-modelling companion [2], the absolute acceleration zero point is

$$\boxed{a_0 = \frac{H_0 c}{2\pi}}. \quad (1)$$

Ontology. The response-theory companion defines the stationary Hysterical response [3]

$$F_H = -\text{Tr}(F \mathcal{L}^{-1} \mathcal{S}) = \int_0^\infty \text{Tr}(F e^{t\mathcal{L}} \mathcal{S}) dt \quad (2)$$

whenever the inverse exists on a stable response subspace. No new messenger appears: F_H is a residual of an already-existing force F . Here F is ordinary gravity; galactic accelerations are that residual organized in space [2]. Horizon packaging of the same residual uses dissipative membrane memory with Euclidean period $\beta = 2\pi/\kappa$ [1, 6, 11].

What is not claimed.

- A modified Newtonian force law or MOND interpolating function.
- A microscopic evaluation of the separate galactic couplings C and A .
- A joint SN/BAO/CMB likelihood for H_0 , or a proof that every galaxy sits exactly on the attractor.

Labeling.

Theorem	Consequence of stated mathematical hypotheses.
Literature hypothesis	Standard geometric/QFT input, cited, not re-proved.
Physical input / assumption	Modelling closure outside pure mathematics.

2 Inherited BTFR structure

The galactic-modelling companion supplies the ballistic residual halo $g_{\text{resp}} = CQ/r$ and, under capacity-filling dynamics, the mature attractor [2]

$$v_f^4 = K M_b, \quad K = \frac{C^2 A^2}{y_*}. \quad (3)$$

Writing the empirical BTFR as $v_f^4 = G a_0 M_b$ [8, 7] identifies

$$G a_0 = K \quad (4)$$

on that attractor. The *exponent* $v_f^4 \propto M_b$ is therefore already theorem material there; the present paper fixes only the absolute scale $a_0 = K/G$.

Assumption 2.1 (Mature attractor normalization). We work on the mature BTFR attractor (3)–(4) of [2], so a single positive acceleration a_0 normalizes all mature response-dominated circular speeds through $v_f^4 = G a_0 M_b$.

3 Horizon memory period

Hysterical Response is history-dependent through the resolvent integral (2). At a causal horizon the same structure is packaged as retarded membrane memory [1].

Literature hypothesis 3.1 (Euclidean horizon period). For a nonextremal Killing horizon with surface gravity $\kappa > 0$, Euclidean regularity requires $\tau \sim \tau + 2\pi/\kappa$, hence a KMS period [6, 11]

$$\beta = \frac{2\pi}{\kappa} \quad (5)$$

(geometric units $\hbar = c = k_B = 1$; c is restored below).

Literature hypothesis 3.2 (Cosmological horizon surface gravity). In de Sitter (or late quasi-de Sitter) geometry the cosmological event horizon has surface gravity $\kappa = H$ [5]. At the present epoch we set $\kappa = H_0$.

Remark 3.3 (Same 2π as black-hole HRT). The horizon-response companion uses precisely (5) to obtain $T_H = \kappa/(2\pi)$ and the Hawking occupation factor [1]. The cosmological application here uses the same geometric period as the memory clock of the residual gravitational channel, not a new fitting constant.

4 IR memory acceleration

Definition 4.1 (IR memory acceleration). Restoring c , the acceleration associated with the horizon memory period (5) is

$$a_{\text{IR}} := \frac{c}{\beta} = \frac{c\kappa}{2\pi}. \quad (6)$$

Theorem 4.2 (Cosmological memory scale). *Under Literature Hypotheses 3.1–3.2 and Definition 4.1 with $\kappa = H_0$,*

$$\boxed{a_{\text{IR}} = \frac{H_0 c}{2\pi}}. \quad (7)$$

Proof. Substitute $\beta = 2\pi/H_0$ into (6). □

Remark 4.3 (Why not cH_0 ?). Raw Hubble time $t_H = 1/H_0$ would give $c/t_H = cH_0$. Horizon regularity replaces t_H by the Euclidean/KMS period $\beta = 2\pi/H_0$, which is smaller by 2π and matches the empirical BTFR scale [8, 7] at $H_0 \sim 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$. Equating Unruh temperature $a/(2\pi)$ to Gibbons–Hawking $H/(2\pi)$ instead yields $a = cH_0$, which is the wrong empirical factor; the HRT reading uses memory time β , not that Unruh identification.

5 Subject independence

Proposition 5.1 (Subject independence of the IR floor). *The scale (7) depends only on the FLRW congruence (through H_0) and on c . It does not depend on the proper age of a particular probe, nor on a passive Hilbert-space relabelling of that probe.*

Remark 5.2. A newly formed grain falls in a preexisting residual potential Φ_H ; it does not reset the channel memory that built Φ_H . Basis covariance of Hysterical Response already forbids passive subject-relabelling from changing F_H [3]. Homogeneity makes H_0 shared by every comoving observer at the present epoch. Local environments (Solar System, dense discs) may have faster gaps and different criticality; they do not invent a second universal cosmological floor [4].

6 Far-halo IR pin

The galactic ballistic halo supports a flat $g_{\text{resp}} \propto 1/r$ regime only inside the propagation length $\lambda = v/\Gamma$ [2]. Extended flat curves therefore require a very small far-halo loss rate Γ .

Assumption 6.1 (Far-halo IR pin). The softest environmentally allowed gap for the residual gravitational channel in an expanding universe is the cosmological-horizon memory rate of Literature Hypotheses 3.1–3.2. The far-halo residual is IR-pinned to that horizon mode, so the characteristic acceleration of the mature residual equals a_{IR} .

Remark 6.2. Assumption 6.1 is the physical bridge from horizon geometry to galactic normalization. It is not a corollary of ballistic transport alone; it is the statement that the soft pole organizing the mature halo is the cosmologically cut-off gravitational residual, not a private galactic clock.

7 Zero-point matching and main theorem

Assumption 7.1 (Zero-point matching). On the mature attractor of Assumption 2.1, the BTFR acceleration equals the IR memory scale:

$$a_0 = a_{\text{IR}}. \quad (8)$$

Equivalently $K = Ga_{\text{IR}}$, fixing the combination $C^2 A^2 / y_*$ without resolving C and A separately.

Theorem 7.2 (BTFR zero point from horizon memory). *Assume Assumptions 2.1, 6.1, and 7.1, together with Theorem 4.2. Then*

$$\boxed{a_0 = \frac{H_0 c}{2\pi}}, \quad v_{\text{f}}^4 = G \frac{H_0 c}{2\pi} M_{\text{b}} \quad \text{on the mature attractor.} \quad (9)$$

Proof. Assumption 7.1 gives $a_0 = a_{\text{IR}}$; Theorem 4.2 gives $a_{\text{IR}} = H_0 c / 2\pi$; Assumption 2.1 with $K = Ga_0$ gives the velocity form. $\square$

Corollary 7.3 (Numerical consistency). *For $H_0 \simeq 67\text{--}70 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [10], equation (9) yields $a_0 \simeq (1.05\text{--}1.12) \times 10^{-10} \text{ m s}^{-2}$, compatible with the gas-rich BTFR calibration $a_0 \simeq 1.20 \times 10^{-10} \text{ m s}^{-2}$ [8, 7] at the precision of the present matching hypotheses.*

8 Relation to MOND phenomenology

MOND/RAR phenomenology uses a universal acceleration a_0 inside an interpolating map $g(g_{\text{b}}, a_0)$ [9]. Hysterical Response uses a different ontology: residual ordinary gravity with ballistic halo $g_{\text{resp}} = CQ/r$ and capacity attractor [2]. Theorem 7.2 explains why an effective a_0 of cosmological size appears, without adopting the MOND force law. Finite-radius mappings therefore need not coincide; only the outer zero point is fixed here.

9 Logical status of the results

Quantity / claim	Status
BTFR attractor $K = C^2 A^2 / y_*$	Inherited from [2].
$\beta = 2\pi / \kappa$	Literature Hypothesis 3.1.
Cosmological $\kappa = H_0$	Literature Hypothesis 3.2.
$a_{\text{IR}} = H_0 c / 2\pi$	Theorem 4.2 (Lean algebra).
Subject independence of a_{IR}	Proposition 5.1.
Far-halo IR pin	Physical Assumption 6.1.
$a_0 = a_{\text{IR}}$	Physical Assumption 7.1.
$a_0 = H_0 c / 2\pi$	Theorem 7.2 (Lean algebra).
MOND interpolating function	Not claimed.
Separate microscopic C, A	Not claimed.

10 Conclusion

The galactic BTFR exponent is settled by capacity dynamics; the absolute zero point is settled by cosmological-horizon Hysterical memory. With Euclidean period $\beta = 2\pi / H_0$ and far-halo IR pinning of the residual gravitational channel,

$$a_0 = \frac{H_0 c}{2\pi}.$$

The 2π is horizon regularity; the subject of the memory is the residual channel on the FLRW congruence, not a particular probe. What remains for phenomenology is coexistence checks against galactic fits and Solar null tests that use this normalization, not a second independent universal acceleration.

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